Energy Transfer

Chapter 4 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION **What evidence shows that energy is transferred from place to place?**

KEY WORDS

energy transfer — Energy transfer means energy moves _____ place to place or object to object.

evidence — Evidence is an observable _____ in the target.

MODEL IT

Energy leaves a source, moves, and changes a target. The change is the evidence of transfer.

SOURCE	→	ENERGY MOVES	→	TARGET CHANGES

SOURCE → PATH → RECEIVER

Sound, light, heat, and electric current can all transfer energy.

SOURCE (WHERE ENERGY STARTS)	PATH (HOW IT TRAVELS)	RECEIVER (WHAT CHANGES)
sound (tapped cup)		
light (flashlight)		
heat (warm water)		
electric current (battery)		

Sound + Light Transfer

Chapter 4 Guided Notes · Page 2

Name: _____ **Date:** _____

SOUND TRANSFERS ENERGY

A tap makes the cup vibrate. The paper target moves even though nothing touched it.

SOURCE	WHAT CHANGED? (EVIDENCE)
tapped cup (sound)	

LIGHT TRANSFERS ENERGY

A flashlight lights a card across the room. A barrier in the path stops the change.

SOURCE	WHAT CHANGED? (EVIDENCE)
flashlight (light)	

DRAW / LABEL

Draw the flashlight, the path of light, and the lit card. Label the SOURCE, PATH, and RECEIVER.

draw and label the light path

Heat + Electric Current Transfer

Chapter 4 Guided Notes · Page 3

Name: _____ **Date:** _____

HEAT TRANSFERS ENERGY

Warm water is placed next to a cooler cup. The temperatures change over time.

BEFORE	LATER	WHAT CHANGED? (EVIDENCE)

ELECTRIC CURRENT TRANSFERS ENERGY

KEY IDEA **Electric current can transfer energy from place to place.**

A battery connects through wires to a bulb, buzzer, or motor. The device changes.

SOURCE	DEVICE	WHAT CHANGED? (EVIDENCE)
battery		

DRAW / LABEL

Draw a battery, wires, and a bulb. Label where the energy moves.

draw and label the circuit

Transfer vs Transformation

Chapter 4 Guided Notes · Page 4

Name: _____ **Date:** _____

TWO DIFFERENT IDEAS

TRANSFER = energy MOVES. **TRANSFORMATION = energy CHANGES FORM.**

A system can show transfer, transformation, or BOTH.

TRANSFER (ENERGY MOVES)	TRANSFORMATION (CHANGES FORM)	BOTH (MOVES AND CHANGES)

CHECK YOUR THINKING

A flashlight: the battery sends energy through wires to the bulb, and the bulb gives off light. Is that transfer, transformation, or BOTH? Explain.

CONNECT *Chapter 1 → 4: energy causes change; now we can trace HOW energy moves to cause that change.*